

Lab 6 Use Case Specifications

Learning objectives

- Translate a system description into complete, verifiable use case specifications.
- Sequence actor actions and matching system responses in a main success scenario.
- Identify triggers, pre/postconditions, and alternative flows for a use case.

Course learning outcome (CLO 2): Elicit, analyze, prioritize and document requirements for a software system.

Use Case Specification

A use case specification is a structured narrative that describes how an actor and the system interact to achieve a goal. It complements the use case diagram from Lab 5 by adding the detail - triggers, conditions, and step-by-step flow of events - needed to implement and test the use case.

Field	Meaning
Use case ID	Short identifier, e.g. UC001, UC002. Use the same ID as in the diagram and inventory.
Use case title	Verb + object name of the use case, taken directly from the diagram.
Brief description	One or two sentences summarizing the actor's goal and the intended outcome.
Primary actors	The actor(s) whose goal the use case serves.
Secondary (supporting) actors	Other people or external systems that provide services needed by the use case. Write 'None' if there are none.
Trigger	The event that starts the use case.
Preconditions	What must already be true before the use case can begin.
Postconditions	What is guaranteed to be true once the use case succeeds.
Main success scenario	Number actor actions and system responses in one sequence (1, 2, 3, ...), placing each step in the appropriate column.
Alternative flows	Numbered exceptions or branches linked to a main-flow step (e.g., 2.a). State where the flow resumes or whether the use case ends.

Part A QU Print Shop - Guided practice

Work through A.1 to A.3 with the instructor, focusing on the two worked specifications.

A.1 Review the use case diagram

Figure 1 is the Print Shop diagram from [Lab 5](#). Part A works through UC001 Submit print job and UC003 Collect printed document. The case describes it as follows:

A user uploads a document and selects the printing options: the number of copies, colour or black-and-white printing, and single- or double-sided printing. Before accepting either a new job or revised options, the Print Shop checks with PAS whether the user has sufficient credit for the resulting printing cost. If sufficient credit is available, the job is accepted and queued for printing; otherwise, the request is rejected and the user is informed. The Print Shop keeps a record of each accepted job, including who submitted it, the selected printing options, and its status: queued, printed or collected.

QU Print Shop Use Case Diagram:

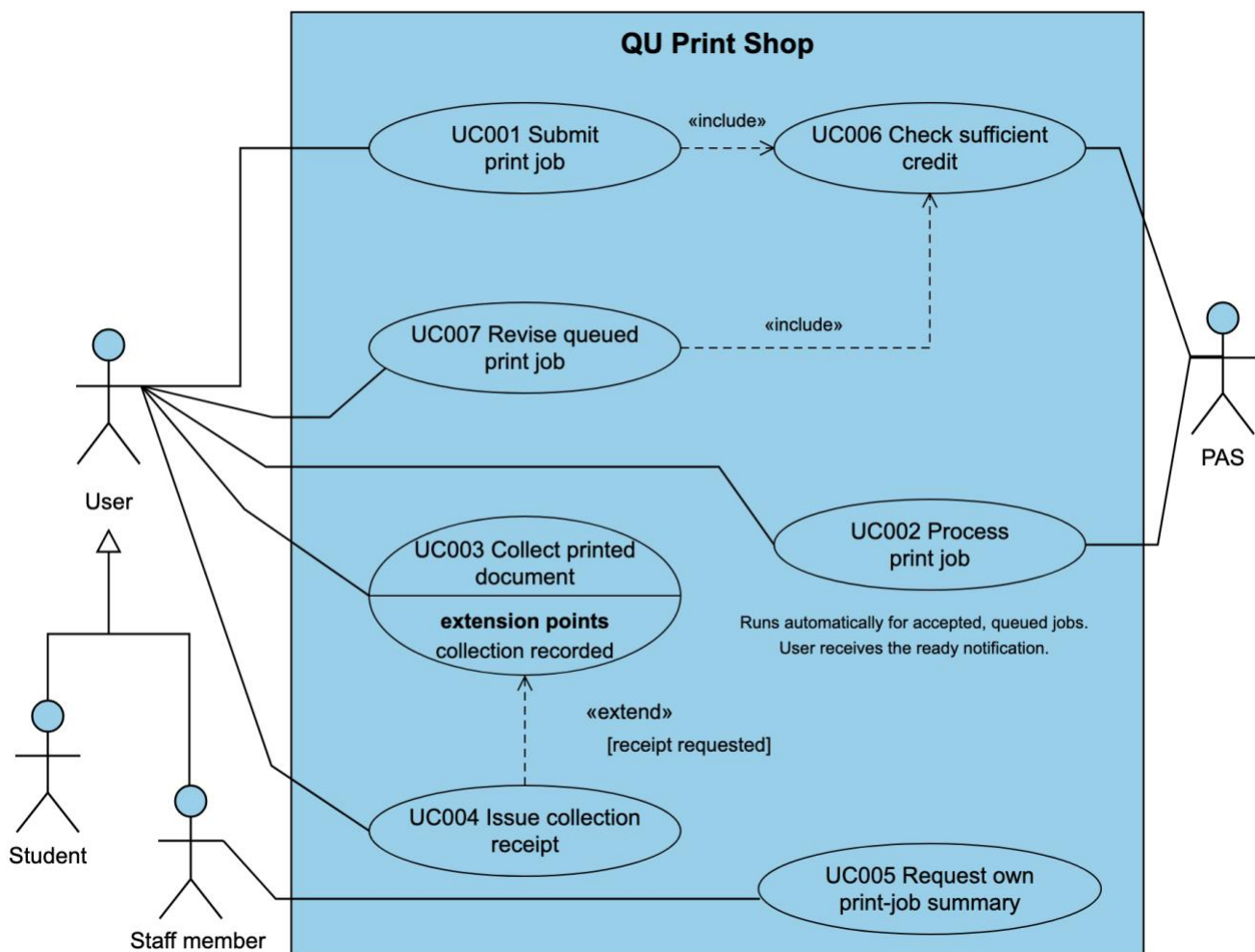


Figure 1: QU Print Shop Use Case Diagram

A.2 Worked example: Submit print job (UC001)

This example is drawn from the QU Print Shop case in Lab 5. Notice how the include relationship (UC001 includes UC006) is documented as a system step, and how step 2.a captures the rejection outcome.

Use case ID:	UC001 Submit print job
Brief description	A user uploads a document and selects printing options. If sufficient credit is confirmed, the Print Shop records the job and queues it for printing.
Primary actors	User (Student, Staff member).
Secondary (supporting) actors	PAS (through UC006 Check sufficient credit).
Trigger	User uploads a document and selects printing options.
Preconditions	The user is a registered student or staff member.
Postconditions	A job for a user with sufficient credit is recorded and queued with its document and options.
Main Success Scenario	
Actor Action	System Response
1. The user uploads a document and selects the number of copies, color or black-and-white printing, and single- or double-sided printing.	2. The Print Shop calculates the printing cost and performs UC006 Check sufficient credit using the user's account and that cost (see 2.a).
	3. The Print Shop records the job with the document and selected options, and queues it for printing.
	4. The Print Shop confirms that the job has been accepted and queued.
Alternative flows:	
2.a. Insufficient credit: if the user's credit is insufficient, the Print Shop rejects the request and informs the user; the use case ends.	

A.3 Worked example: Collect printed document (UC003)

This example documents the extend relationship from Lab 5 (UC004 extends UC003). The base names the extension point at step 2. UC004 states the extension point and condition in its Trigger; collection remains complete without a receipt.

Use case ID:	UC003 Collect printed document
Brief description	The user collects a printed document, and the Print Shop records its collection.
Primary actors	User (Student, Staff member).
Secondary (supporting) actors	None.
Trigger	User submits a collection request for a printed job.
Preconditions	The job status is "printed".
Postconditions	The job is marked "collected".
Main Success Scenario	
Actor Action	System Response
1. The user submits a collection request for the printed job.	2. The Print Shop records the job as collected. [Extension point: collection recorded]
	3. The Print Shop confirms that collection has been recorded.
Alternative flows:	
None specified.	

Collection can finish without a receipt. UC004 extends UC003 only at the named extension point when a receipt is requested.

In the separate UC004 specification, write its **Trigger** as: "During UC003 Collect printed document, at extension point *collection recorded*, if a receipt has been requested." Only the two base specifications are worked through in this lab.

Part B All Phones Shop - Individual specifications

The instructor will review the Lab 5 use case diagram solution in Figure 2 with you, followed by the Register client example (15 minutes). For the individual tasks, use this solution and the APS case from Lab 3. You may discuss ideas, but write your own specifications in the supplied template.

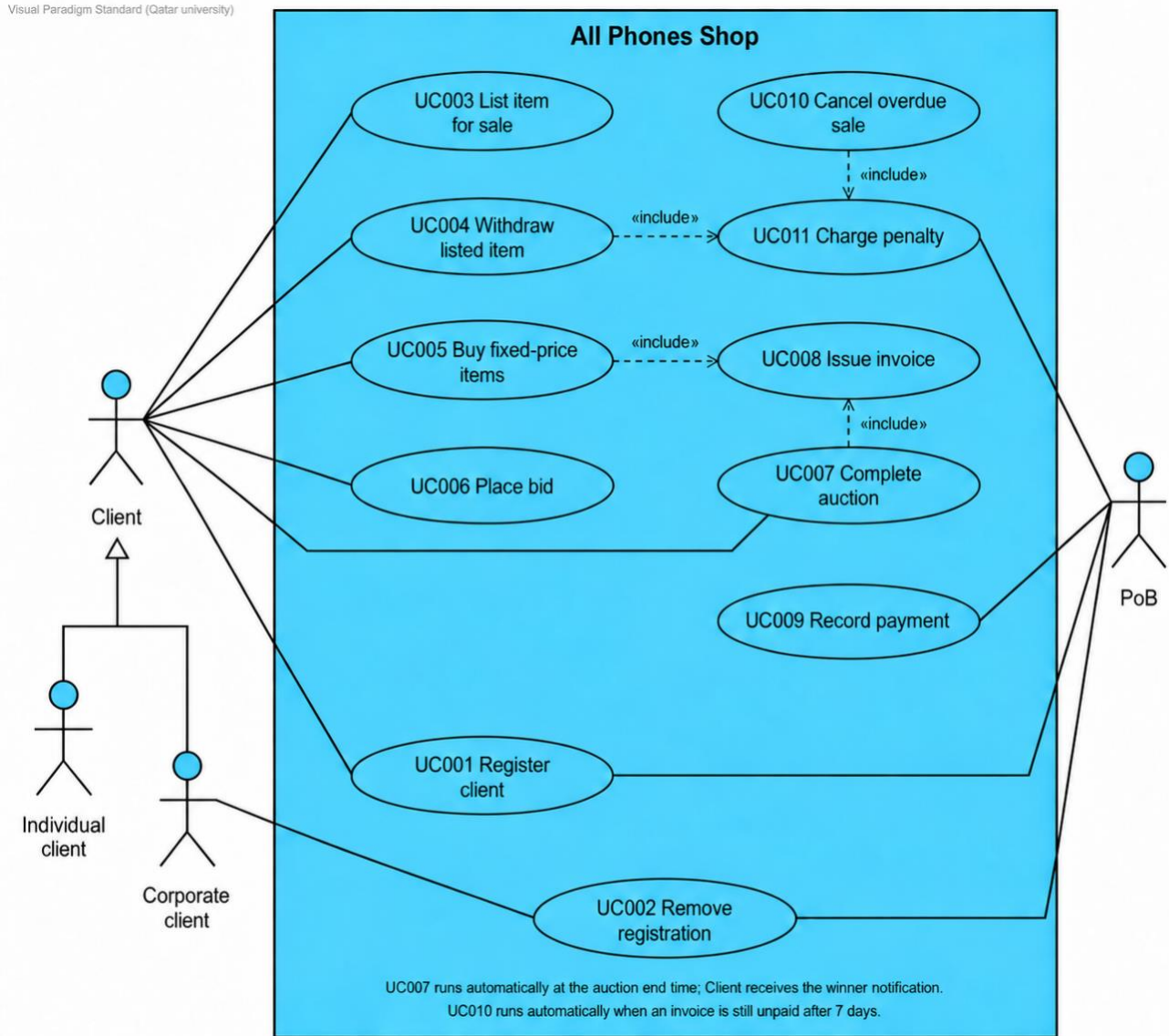


Figure 2: All Phones Shop Use Case Diagram

Worked example: Register client - 15 minutes

Use case ID:	UC001 Register client
Brief description	The client creates a new registration with APS and PoB is informed about the new client.
Primary actors	Client.
Secondary (supporting) actors	PoB.
Trigger	The client requests registration by providing a name and address.
Preconditions	Any required login checks have been completed outside APS.
Postconditions	A registration with a unique number exists. PoB has been notified of that number and has opened the corresponding financial account.

Main Success Scenario	
Actor Action	System Response
1. The client provides name and address.	2. APS checks whether the client already has a registration (see 2.a).
	3. APS creates the registration with the client's details and assigns a unique registration number.
	4. APS sends the assigned registration number to PoB, which opens the client's financial account under that number.
	5. APS informs the client of the registration number.
Alternative flows: 2.a. Existing client: If the client already exists, inform the client that a new registration cannot be created. The use case ends without creating another registration.	

Task 1: List item for sale (UC003) - 30 minutes

Develop the specification for UC003 List item for sale, using Figure 2 and the APS case from Lab 3. Include image approval and regeneration within this use case. Add rows to the template as needed.

Use case ID:	UC?? <Use case title>
Brief description	
Primary actors	
Secondary (supporting) actors	
Trigger	
Preconditions	
Postconditions	
Main Success Scenario	
Actor Action	System Response

Alternative flows:

?.a.

Task 2: Place bid (UC006) - 30 minutes

Develop the specification for UC006 Place bid, using Figure 2 and the APS case from Lab 3. End with the bid accepted and recorded, or rejected. Auction completion and invoicing belong to UC007 Complete auction and are outside this specification. Add rows as needed.

Use case ID:	UC?? <Use case title>
Brief description	
Primary actors	
Secondary (supporting) actors	
Trigger	
Preconditions	
Postconditions	
Main Success Scenario	
Actor Action	System Response
Alternative flows:	

Submission requirements

15 minutes. Review both specifications, then open the [CMPS310 GitHub repositories](#) and select your own repository. Submit **lab06-solution.docx** inside a folder named **Lab 6 Use Case Specification**, using the individual repository workflow from Lab 4. Part A is not submitted. Commit and push the file, then check that it opens from your repository.

File	Required contents
lab06-solution.docx	Name and student ID; completed specifications for UC003 List item for sale and UC006 Place bid; any stated assumptions.

Assessment criteria (10 points)

Criterion	Points
Correct and complete template fields (ID, title, description, actors, trigger, pre/postconditions)	3
Main success scenario correctly sequenced with matching actor actions and system responses	3
Alternative flows correctly identified, numbered and referenced from the step they extend	2
Consistency with the Lab 5 use case diagram and the APS case study	1
Clear, professional notation and writing	1

Further reading

[Visual Paradigm - What is Use Case Specification](#)

[OMG UML 2.5.1 Section 18 Use Cases](#)